Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Lesson 1: Model Selection Techniques

- Introduction to Model Selection
  Reading • 5 min
- Bayesian Model Selection
  Reading • 30 min
- Bayesian Model Selection Readings
  Reading • 1h

Lesson 2: Cross Validation

- Introduction to Cross-Validation
  Reading • 25 min
- K-Fold Cross-Validation
  Reading • 2 min
- Leave-One-Out Cross-Validation (LOOCV)
  Reading • 10 min

Lesson 3: Introduction to Optimization Techniques

- Introduction to Optimization Techniques
  Reading • 13 min
- Optimization Using Gradient Descent
  Reading • 40 min
- Gradient Descent with Momentum
  Reading • 45 min
- Stochastic Gradient Descent (SGD)
  Reading • 40 min
- Constrained Optimization
  Reading • 16 min
- Lagrange Multipliers
  Reading • 8 min
- **Convex Optimization**
  Reading • 8 min
- Linear Programming
  Reading • 15 min
- Quadratic Programming
  Reading • 10 min
- Check Your Knowledge: Optimization Techniques
  Graded Assignment • 15 min

Convex Optimization

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In optimization, we often deal with problems where we aim to find the minimum or maximum of a function, subject to certain constraints. Convex optimization is a particularly useful class of optimization problems where the objective function and the constraints are convex. The significance of convex optimization lies in its guarantee of global optimality—if we find a local minimum, it is also the global minimum.

Convex Sets and Convex Functions

Convex Sets:

- A set C is convex if, for any two points x and y in C, the line segment connecting x and y lies entirely within C. Mathematically, this is expressed as:

$$\theta x + (1-\theta)y \in C \text{ for all } \theta \in [0,1]$$

- Convex sets are important because they define the feasible region in a convex optimization problem. If the feasible region is convex, any local minimum within this region is also the global minimum.

Convex Functions:

- A function $f(x)$ is convex if, for any two points x and y in its domain, and for any $\theta \in [0,1]$, the following holds:

$$f(\theta x + (1-\theta)y) \leq \theta f(x) + (1-\theta)f(y)$$

- This property implies that the graph of a convex function lies below the straight line connecting any two points on the graph. Convex functions ensure the optimization landscape has no local minima other than the global minimum.

Properties and Examples

Strong Duality:

- In convex optimization, strong duality typically holds, meaning the optimal value of the dual problem is equal to the optimal value of the primal problem. This property simplifies solving optimization problems by allowing us to solve either the primal or the dual problem.

Epigraph:

- The epigraph of a convex function is the set of points lying on or above the graph of the function. Mathematically, the epigraph of $f(x)$ is defined as:

$$\text{epi(f)} = \{(\text{x}, \text{t}) \in \mathbf{R}\text{^D} \times \mathbf{R} \mid \text{f(x)} \leq \text{t}\}$$

- The epigraph is a convex set if $f(x)$ is a convex function, providing a geometric interpretation of convexity.

Convexity and the Gradient:

- A differentiable function $f(x)$ is convex if, for any points x and y, the following holds:

$$f(y) \geq f(x) + \nabla f(x)\text{^}T(y-x)$$

- This expression means the function lies above its linear approximation, which is a defining characteristic of convex functions.

Hessian and Convexity:

- If a function $f(x)$ is twice differentiable, it is convex if and only if its Hessian matrix $\nabla^2 f(x)$ is positive semi-definite for all x in its domain.

Closure Properties:

- Convex functions have important closure properties. For instance, the nonnegative weighted sum of convex functions is also convex. This means that if $f_1(x)$ and $f_2(x)$ are convex functions, then any non-negative linear combination $\alpha f1(x) + \beta f2(x)$ (with $\alpha, \beta \geq 0$) is also convex.

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